

From Data Warehouses to Lakehouses: Unifying Data Management & Analytics

Based on papers:

“Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics” &

“The Data Lakehouse: DataWarehousing and More”

Submitted By:

Saksham Chauhan

M.Tech. in Artificial Intelligence and Data Science

Indian Institute of Information Technology, Kottayam, Kerala, India

Content

- Introduction to Data Systems (Introduction to Research Area)
- OTLP vs OLAP
- Data Warehousing
- Data Lake
- Why the two-tier architecture fails? (Problem Statement)
- Table Formats (Methodology)
- Data Lakehouse
- Conclusion & Future Scope
- References

Introduction to Data Systems

- Rapid growth of structured, semi-structured and unstructured data
- Increasing demand for analytics, BI and ML
- Requirements of an efficient data system:
 - Scalable storage
 - Efficient Processing
 - Unified data access across systems
- Traditional Systems:
 - Online Transaction Processing (OLTP) for operational delivery
 - Online Analytical Processing (OLAP) for analytics delivery

OLTP VS OLAP: Operational v/s Analytics Systems

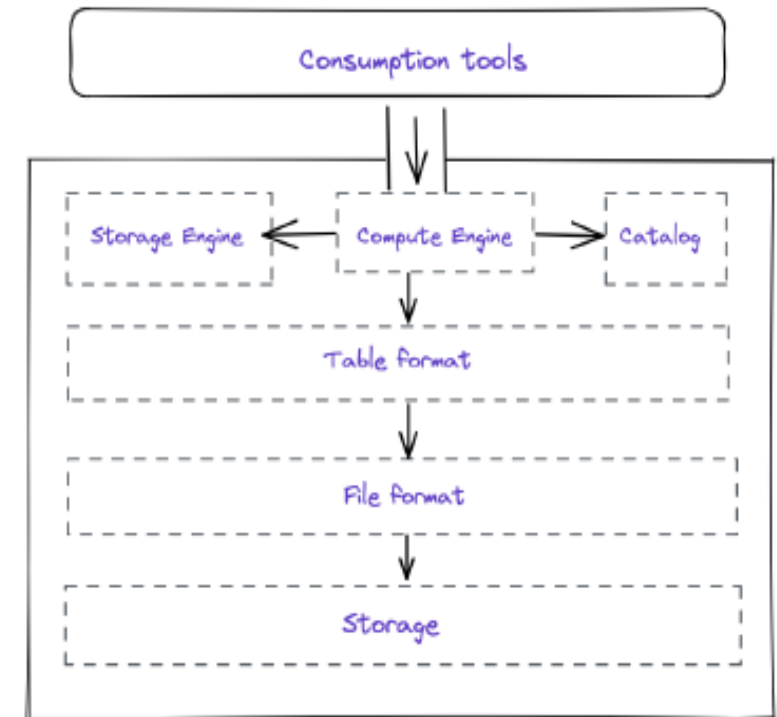
- RDBMS-OLTP
 - Designed for real-time transactional processing
 - Handles CRUD on small data volumes
 - Optimized for row-level access
- RDBMS-OLAP / Data Warehouses
 - Designed for large scale processing and analytical queries
 - Uses columnar storage for efficiency
 - Integrated data from multiple sources such as CRM, application db's, etc.

Data Warehousing (RDBMS-OLAP Systems)



Data Platform Architecture for a Data Warehousing System

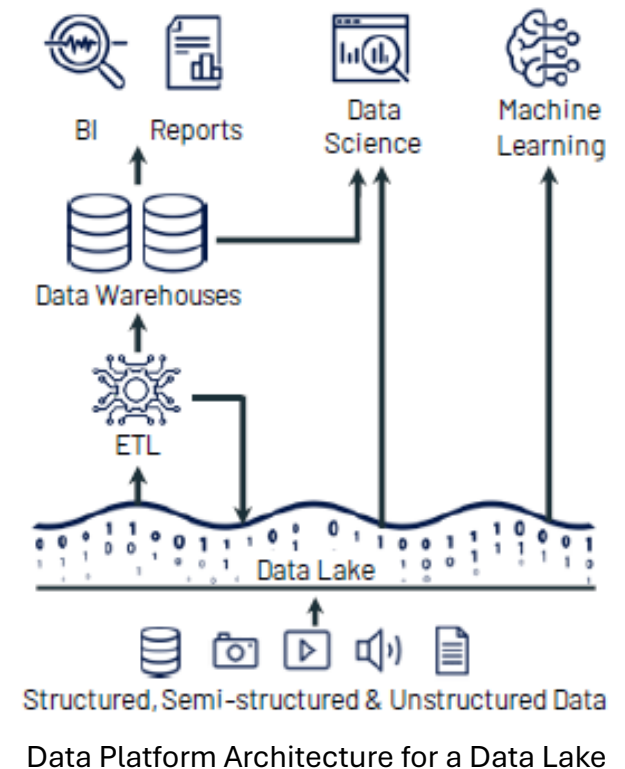
- Technical Components
 - Data ingestion from multiple sources
 - ETL pipelines
 - Storage using columnar formats
 - SQL-based querying engines & BI tools
- Technical Capabilities
 - Governance & Security
 - High concurrency
 - Low Query Latency
 - Workload Management
 - ACID Compliance
- Technical Limitations
 - Limited to Structured Workloads
 - High storage and infrastructure costs
 - Schema-on-write
 - Complex ETL pipelines



A generic representation of an RDBMS-OLAP data warehouse

Data Lake: Storage for Diverse Data

- What it is
 - A scalable storage system for structured, semi-structured & unstructured data
 - Uses low-cost storage objects (Amazon S3, HDFS)
 - Often a two-tier architecture = Storage + Compute layer
- Strength
 - Cost effective compared to Data Warehousing
 - Follows Schema-on-Read approach
 - Suitable for ML, BI & Analytics
- Limitations
 - Lack of ACID transaction compliance
 - No schema enforcement
 - Often leads to 'data-swamp'
 - ETL into Lake, ELT into Warehouse



Why the Two-Tier Architecture Fails?



Separation of data storage and compute layer leads to increased latency, staleness, data duplication, data mismanagement and data inconsistencies.

Table Formats: Enabling Reliable Data Lakes

- What it is
 - A data management layer built on top of data lake storage
 - Ex: Apache Iceberg, Delta Lake, Apache Hudi
- Features
 - Metadata management
 - Schema enforcement
 - Versioning & time travel
 - Support for updates, merges & deletes
 - ACID compliant data updates

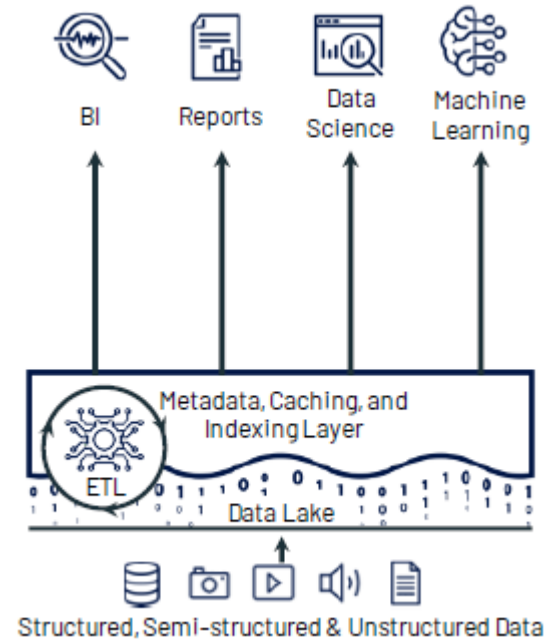
Data Lakehouse

“A data management system based on low-cost and directly-accessible storage that also provides traditional analytical DBMS management and performance features such as ACID transactions, data versioning, auditing, indexing, caching, and query optimization”.

- Technical Components (an implementation example)
 - Data Storage: Data Lake (ex. S3)
 - Storage Engine: Apache Spark
 - File Format: Open Formats (ex. Parquet)
 - Table Format: Apache Hudi / Delta Lake
 - Catalog: HDFS / Hive
 - Compute Engine: Apache Spark (for ML) / Apache Flink (for streaming)

Additional Benefits of Data Lakehouses

- Governance & Security
- High Concurrency
- Low Query Latency
- Ad-Hoc Queries
- Workload Management
- ACID-complaint transactions
- Separation of Storage & Compute
- Open Data Architecture



Data Platform Architecture for a Data Lakehouse

Conclusion and Future Scope

- **Conclusion:**

- Traditional data warehouses provide high-performance analytics but lack flexibility
- Data lakes offer scalability and flexibility but suffer from poor data reliability
- Table formats introduce a data management layer that ensures consistency and performance
- Data lakehouse architecture unifies data warehousing and advanced analytics in a single system

- **Future Scope:**

- Wider industry adoption of lakehouse architectures
- Better integration of machine learning and analytics workflows
- Standardization of table formats across platforms
- Enhanced query optimization and performance improvements

References

- Armbrust, M., et al. (2021). *Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics*. CIDR.
- Gao, et al. (2023). *The Data Lakehouse: Data Warehousing and More*. arXiv preprint.



Thank You!

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